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Inventor(s)	RJ; Yogesh et al.

KNOB MECHANISM

Abstract

A knob mechanism according to the disclosure may include a knob body and a lock mechanism associated with the knob body. The lock mechanism may include a locking member that is movable between an extended position and a retracted position, a pivot member that is pivotable with respect to the knob body to move the locking member, and a button that is operable to pivot the pivot member to move the locking member from the extended position toward the retracted position.

Inventors: **RJ; Yogesh (Bengaluru, IN), MUTHUSWAMY; Sendil (Bengaluru, IN), MURALIDHARA; Prathap (Bengaluru, IN), RAVISHANKAR; Ankith (Bengaluru, IN), SUNNY; Charles (Bengaluru, IN), PETER; Abin (Bengaluru, IN)**

Applicant: **TEREX SOUTH DAKOTA, INC.** (Watertown, SD)

Family ID: **96881154**

Assignee: **TEREX SOUTH DAKOTA, INC. (Watertown, SD)**

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Background/Summary

TECHNICAL FIELD

[0001] The disclosure relates to a knob mechanism.

BACKGROUND

[0002] A prior crank handle including a knob for a lift device is disclosed in International Publication No. WO 2013/140149 A1.

SUMMARY

[0003] A knob mechanism according to the disclosure may include a knob body and a lock mechanism associated with the knob body. The lock mechanism may include a locking member that is movable between an extended position and a retracted position, a pivot member that is pivotable with respect to the knob body to move the locking member, and a button that is operable to pivot the pivot member to move the locking member from the extended position toward the retracted position.

[0004] While exemplary embodiments are illustrated and disclosed, such disclosure should not be construed to limit the claims. It is anticipated that various modifications and alternative designs may be made without departing from the scope of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. **1** is a perspective view of an example lift including a support member arrangement shown in a lowered position, and a drive system for moving the support member arrangement, wherein the drive system includes a knob mechanism according to disclosure;

[0006] FIG. **2** is a perspective view of the lift of FIG. **1** with the support member arrangement shown in a raised position;

[0007] FIG. **3**A is a front view of the support member arrangement including first, second and third upright support members, wherein the second upright support member is shown in section and the third upright support member is partially broken away to show first and second drive members of the drive system positioned inside of the second and third upright support members, respectively;

[0008] FIG. **3**B is a top view of the support member arrangement and drive system;

[0009] FIG. **4** is a schematic diagram of the support member arrangement;

[0010] FIG. **5** is an enlarged front view of a portion of the drive system including the knob mechanism; and

[0011] FIG. **6** is an enlarged perspective view of the knob mechanism;

[0012] FIG. **7** is a side cross-sectional view of the knob mechanism in an extended or locked position;

[0013] FIG. **8** is a side cross-sectional view of the knob mechanism in a retracted position;

[0014] FIG. **9** is a side view of the knob mechanism mounted on a rotatable member of the drive system;

[0015] FIG. **10** is an exploded perspective view of the knob mechanism;

[0016] FIG. **11** is a perspective view of another embodiment of a knob mechanism according to the disclosure;

[0017] FIG. **12** is an exploded view of the knob mechanism shown in FIG. **11**;

[0018] FIG. **13** is a side cross-sectional view of the knob mechanism of FIG. **11** shown in an extended or locked position;

[0019] FIG. **14** is a side cross-sectional view of the knob mechanism of FIG. **11** in a retracted position;

[0020] FIG. **15** is a perspective view of yet another embodiment of a knob mechanism according to the disclosure;

[0021] FIG. **16** is a side cross-sectional view of the knob mechanism of FIG. **15** shown in an extended or locked position; and

[0022] FIG. **17** is a side cross-sectional view of the knob mechanism of FIG. **15** in a retracted position.

DETAILED DESCRIPTION

[0023] As required, detailed embodiments are disclosed herein. However, it is to be understood that the disclosed embodiments are merely exemplary, and that various and alternative forms may be employed. The figures are not necessarily to scale; some features may be exaggerated or minimized to show details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art.

[0024] As used in the description of the various described embodiments and the appended claims, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms “includes,” “including,” “comprises” and/or “comprising” specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0025] FIGS. **1** and **2** show an example lift device or lift **10**, such as a boom lift or aerial work platform. The lift **10** includes a base **12**, a support member arrangement **14** attached to the base **12**, a drive system **16** that is operable to move the support member arrangement **14** between a lowered position, shown in FIG. **1**, and a raised position, shown in FIG. **2**, and a work platform assembly **18** connected to the support member arrangement **14** and configured to receive an operator. As explained below, the drive system **16** includes a novel knob mechanism **20** (e.g., handle mechanism) according to the disclosure, and the knob mechanism **20** may function as a locking mechanism for locking the support member arrangement **14** in a desired position. As another example, the knob mechanism **20** may be used in any other suitable application (e.g., device) in which it may be desirable to lock one component (e.g., rotatable component) with respect to another component.

[0026] The base **12** of the lift **10** shown in FIGS. **1** and **2** is configured to support the support member arrangement **14**, along with the rest of the lift **10**, and includes a main body **22**, such as a rigid support structure, and one or more support members **24**, such as wheels, casters, tracks, outriggers, etc., attached to the main body **22**. In the illustrated embodiment, the lift **10** includes multiple support members **24** formed as wheels and associated tires, so that the base **12** is movable. Furthermore, two or more of the wheels are pivotable in order to steer the base **12**.

[0027] The support member arrangement **14** may include one or more movable support members **26**, such as pivotable and/or linearly movable booms, masts, tubes, columns, arms, etc., attached to the base **12**. In the illustrated embodiment, the support member arrangement **14** includes first, second and third upright support members **26***a*, **26***b* and **26***c*, respectively, and the second and third upright support members **26***b* and **26***c*, respectively, are linearly movable with respect to the first upright support member **26***a* when the support member arrangement **14** moves between the lowered and raise positions. In another embodiment, the support member arrangement **14** may include two upright support members, wherein one of the upright support members may be movable with respect to the other upright support member.

[0028] Referring to FIGS. **1-4**, the drive system **16** may have any suitable configuration that is operable to move one or more of the support members **26** in order to move the support member arrangement **14** between the lowered and raised positions. For example, the drive system **16** may include one or more drive members **28**, such as flexible drive members (e.g., strands, belts, chains and/or cables that may be arranged as one or more pulley mechanisms, for example), rigid drive members (e.g., rack and pinion systems, rotatable shafts and gears), etc., and one or more actuators, such as a manually operable (i.e., hand-operated) actuator **30** for moving the drive members **28**. In the illustrated embodiment, the drive system **16** includes movable first and second flexible drive members **28***a* and **28***b*, respectively, which may be lower and upper drive members, respectively. Furthermore, each drive member **28***a*, **28***b* may have any suitable form or configuration, such as a strand, belt, chain, cable, or other flexible member. As a more detailed example, each drive member **28***a*, **28***b* may be a continuous drive member that is movable (e.g., circulatable or reciprocatable) with respect to one or more rotatable members **32**, such as pulleys, rollers, wheels, sheaves, etc. In the illustrated embodiment, the first drive member **28***a* is movable about or at least partially around two rotatable members **32** attached at opposite ends of the second upright support member **26***b* (e.g., mounted on an interior surface of the second upright support member **26***b*), and the second drive member **28***b* is movable about or at least partially around two rotatable members **32** attached at opposite ends of the third upright support member **26***c* (e.g., mounted on an interior surface of the third upright support member **26***c*), so that a portion of each drive member **28***a*, **28***b* may reciprocate (e.g., move back and forth) between the respective two rotatable members **32**. In the illustrated embodiment, the drive system **16** also includes an idler rotatable member **34**, such as a pulley, roller, wheel, sheave, etc., for the second drive member **28***b* that is positioned between the rotatable members **32** associated with the second drive member **28***b*. In addition, in the embodiment shown in FIGS. **3**A and **3**B, the first drive member **28***a* is positioned or extends between the second upright support member **26***b* and the first upright support member **26***a* (e.g., between the interior surface of the second upright support member **26***b* and an exterior surface, such as a front face, of the first upright support member **26***a*), and the second drive member **28***b* is positioned or extends between the third upright support member **26***c* and the second upright support member **26***b* (e.g., between the interior surface of the third upright support member **26***c* and an exterior surface, such as a front face, of the second upright support member **26***b*).

[0029] In the illustrated embodiment, the first drive member **28***a* is movably received or positioned within the second upright support member **26***b* and fixedly connected to the first upright support member **26***a* at a fixed connection location **36***a* (e.g., within the second upright support member **26***b*) and to the third upright support member **26***c* at a fixed connection location **36***b* (e.g., within the third upright support member **26***c* or on an exterior of the third upright support member **26***c*), and the second drive member **28***b* is movably received or positioned within the third upright support member **26***c* and fixedly connected to the second upright support member **26***b* at a fixed connection location **38** (e.g., within the third upright support member **26***c*). With such a configuration, load from the third upright support member **26***c* may be transferred to one side of the first drive member **28***a*, and force or tension in the second drive member **28***b* resulting from the actuator **30** may be transferred to the second support member **26***b*. In addition, the first and second drive members **28***a* and **28***b*, respectively, are configured to cooperate to provide synchronized movement of the second and third upright support members upon actuation by the actuator **30**.

[0030] The drive members **28***a*, **28***b* may be fixedly connected to the upright support members **26***a*, **26***b* and/or **26***c* in any suitable manner, such as with any suitable connectors or fasteners (e.g., clamps, brackets, screws, bolts and nuts, etc.). In the illustrated embodiment, a clamp **40** is used for fixedly connecting the first drive member **28***a* to each of the first upright support member **26***a* and the third upright support **26***c*, and for fixedly connecting the second drive member **28***b* to the second upright support member **26***b*. Further details of an example clamp, as well as further details of an example lift with which the knob mechanism **20** may be used, are disclosed in U.S. application Ser. No. 18/593,237 (corresponding to Attorney Docket TERS 0147 PUS) filed on Mar. 1, 2024, which is hereby incorporated by reference in its entirety.

[0031] Returning to FIG. **3**A, the clamp **40** at location **36***a* may be fixedly attached to the exterior surface, e.g., the front face, of the first upright support member **26***a*. The clamp **40** at location **36***b* may be fixedly attached to a bracket or bracket arrangement **42** (e.g., first and second bracket portions **42**_a_ and **42**_b_, respectively) that extends through the second upright support member **26**_b_ and is fixedly attached to the third upright support member **26**_c_. For example, the bracket **42** may extend through a longitudinally extending slot or channel **44** (shown in FIG. **2**) formed in a side of the second upright support member **26**_b_. With such a configuration, the bracket **42** may move along the channel **44** when the support member arrangement **14** moves between the lowered and raised positions. The clamp **40** at location **38** may be fixedly attached to the exterior surface, e.g., the front face, of the second upright support member **26**_b_.

[0032] As also shown in FIG. **3**A, the lift **10** may include one or more additional actuators associated with the support member arrangement **14** and configured to assist with movement of the support member arrangement **14**. In the illustrated embodiment, the lift **10** includes an additional or second actuator **45**, such as a cylinder or gas spring, or any suitable biasing member or spring, having one end connected to the base **12** and an opposite end connected to the second upright support member **26**_b_ (e.g., upper end of the second upright support member **26**_b_). With such a configuration, the second actuator **45** may also act as a counterweight, for example,

[0033] Referring to FIGS. **1**-**3**B, **5** and **6**, the actuator **30** includes a rotatable member **46**, such as a wheel, disc, hand crank, lever arm, etc., that is configured to move one or more of the drive members **28**_a_, **28**_b_, such as through a transmission **47**, and the knob mechanism **20** mounted (e.g., rotatably mounted) on the rotatable member **46**. The rotatable member **46** may be mounted on one of the support members **26** and/or on the work platform assembly **18**, for example. In the illustrated embodiment, the rotatable member **46** comprises a lever arm mounted on a shaft **48** that is rotatably mounted on the third upright support member **26**_c_, and the transmission **47** includes a gear arrangement **50** that cooperates with the lever arm **46** to transfer force from the lever arm **46** to the second drive member **28**_b_. For example, the gear arrangement **50** may include a first gear **52** at or near an end of the lever arm **46** and that is rotatable with the lever arm **46**, and a second gear **54** that is configured to mesh with the first gear **52**. In the embodiment shown in FIGS. **3**A, **3**B and **5**, the first gear **52** is fixedly mounted on the shaft **48** to which the lever arm **46** is fixedly mounted, the second gear **54** is connected to a rotatable output shaft **56** that extends through a body of the third upright support member **26**_c_, and the output shaft **56** is fixedly connected to a rotatable drive element **58**, such as a pulley, ring, wheel, sheave, etc., that is configured to drive the second drive member **28**_b_ upon rotation of the lever arm **46**. As explained below, the knob mechanism **20** may be configured to lock the lever arm **46** in a desired position in order to lock the support member arrangement **14** and work platform assembly **18** in a corresponding desired position, or otherwise inhibit or prevent movement of the support member arrangement **14**.

[0034] Returning to FIG. **1**, the work platform assembly **18** is movable with the support member arrangement **14** when the support member arrangement **14** moves between the lowered and raised positions. Specifically, in the illustrated embodiment, the work platform assembly **18** is fixedly mounted on the third upright support member **26**_c_. The work platform assembly **18** includes a base or work platform **60** (e.g., flat work platform) sized to receive and support an operator thereon, and a frame assembly **62**, such as a railing assembly, provided on the platform **60**. In the illustrated embodiment, the frame assembly **62** extends around, or partially around, a perimeter of the platform **60**, and includes multiple frame members, such as toe boards **64**, horizontal rails **66**, and vertical or upright rails **68**, which are connected to the toe boards **64** and the horizontal rails **66**. The frame assembly **62** further includes one or more movable (e.g., pivotable) doors **69** to allow operator ingress to, and egress from, the work platform assembly **18**.

[0035] Referring to FIGS. **5**-**8**, the actuator **30** including the knob mechanism **20** will now be described in more detail. The knob mechanism **20** may be configured to cooperate with a locking member, such as a locking plate, block, bar, etc., to lock the lever arm **46** in one or more desired positions. In the illustrated embodiment, the locking member is formed as a locking plate **70** connected to the third upright support member **26**_c_ or the work platform assembly **18** and having multiple locking features, such as locking apertures **72**, and the knob mechanism **20** is engageable or otherwise cooperable with any of the apertures **72** in order to lock the lever arm **46** in any one of multiple desired use positions. Specifically, the knob mechanism **20** includes a knob body **74**, such as a casing or housing (e.g., curved body, rounded body, rectangular body, etc.), and a lock mechanism **76** associated with the knob body **74**. The lock mechanism **76** includes a locking member **78**, such as a pin, rod, shaft, column, etc., that is movable between a first or extended position (e.g., locked position or non-actuated position) and a second or retracted position (e.g., released position or actuated position), a pivot member **80**, such as a lever, arm, etc., that is pivotable with respect to the knob body **74** to move the locking member **78**, and a button **82** that is operable to pivot the pivot member **80** to move the locking member **78** from the extended position toward the retracted position. Referring to FIGS. **1** and **2**, the actuator **30** may also be provided with a cover **83**, such as a rotatable cover, for covering a main body of the lever arm **46** and the locking plate **70**.

[0036] In the embodiment shown in FIGS. **7** and **8**, the locking member **78** comprises an elongated locking pin **84** having a longitudinal axis **86**, and an enlarged locking portion or end **88** attached to the locking pin **84**. For example, the locking portion **88** may be attached to the locking pin **84** with a fastener **90** (e.g., connecting pin, bolt, screw, etc.) that extends in a direction transverse to the longitudinal axis **86** of the locking pin **84**, or the locking portion **88** may be integrally formed with the locking pin **84** (e.g., the locking portion **88** and the locking pin **84** may be formed as one piece). Furthermore, the pivot member **80** may be engageable with the locking member **78** to move the locking member **78** from the extended position toward the retracted position. In the illustrated embodiment, for example, the pivot member **80** is disposed in an interior of the knob body **74** and has a curved first end **92** that is engageable with an enlarged head **94** of the locking pin **84** when the locking member **78** is in the retracted position shown in FIG. **8**. In addition, the button **82** may be slidably mounted on the knob body **74** and engageable with the pivot member **80** to pivot the pivot member **80**. For example, the pivot member **80** may have an opposite curved second end **96** that is directly engageable with the button **82** when the locking member **78** is in the extended position shown in FIG. **7**. In the embodiment shown in FIGS. **7** and **8**, the locking member **78** is movable (e.g., slidable) in a first direction **98** (e.g., parallel to the longitudinal axis **86**), and the button **82** is movable (e.g., slidable) in a second direction **100** transverse to the first direction **98** in order to move the locking member **78** in the first direction **98**. More specifically, the button **82** is movable in the second direction **100** from a first position (e.g., extended position) shown FIG. **7** to a second position (e.g., depressed position) shown in FIG. **8** in order to pivot the pivot member **80** and thereby move the locking member **78** from the extended position shown in FIG. **7** to the retracted position shown in **8**. In the illustrated embodiment, the second direction **100** is perpendicular to the first direction **98**. In addition, the knob body **74** may cooperate with the button **82** to guide movement of the button **82** between the first and second positions. For example, the knob body **74** may define an opening having parallel sidewalls that are configured to slidably receive parallel sidewalls of the button **82**. With the above configuration, the button **82** may be easily and efficiently actuated when an operator grips the knob body **74**.

[0037] Referring to FIG. **7**, the pivot member **80** may be offset with respect to the second direction **100** when the locking member **78** is in the extended position, in order to facilitate pivotal movement of the pivot member **80** when the button **82** is moved from the first position toward the second position. For example, the pivot member **80** may have a longitudinal axis **102** that extends at an angle greater than 0° (e.g., an angle in the range of 5 to 10°) with respect to the second direction **100** when the locking member **78** is in the extended position.

[0038] Referring to FIG. **8**, the pivot member **80** may also be offset with respect to the first direction **98** when the locking member **78** is in the retracted position. For example, longitudinal axis **102** of the pivot member **80** may extend at an angle greater than 0° (e.g., an angle in the range of 18 to 25°) with respect to the first direction **98** when the locking member **78** is in the retracted position.

[0039] While the knob body **74** and the locking mechanism **76** may be associated with each other in any suitable manner, FIGS. **6-10** show an example configuration for connecting together the knob body **74** and the locking mechanism **76**. In the illustrated embodiment, the knob body **74** is attached to a mounting bracket **104** along with an additional bracket or pivot member bracket **106** such that the pivot member bracket **106** is disposed in the interior of the knob body **74**, and the knob mechanism **20** further includes a housing **108** that is also connected to the mounting bracket **104** and configured to receive the locking portion **88** of the locking member **78**. The knob body **74**, the pivot member bracket **106** and the housing **108** may each be connected to the mounting bracket **104** in any suitable manner, such as with one or more fasteners (e.g., screws, bolts, nuts, etc.). In the illustrated embodiment, the knob body **74** and the pivot member bracket **106** are attached to the mounting bracket **104** with two screws or bolts **110** and corresponding nuts **112**, and the housing **108** is connected to the mounting bracket **104** with two screws **114** so that the mounting bracket **104** is positioned between the knob body **74** and the housing **108**. Furthermore, the pivot member **80** may be pivotally attached to the pivot member bracket **106** in any suitable manner, such as with a pivot pin **116**, so that the pivot member **80** is pivotable with respect to the knob body **74** about a pivot axis that may be positioned within the interior of the knob body **74**. In the embodiment shown in FIGS. **7** and **8**, the pivot axis extends between the button **82** and the locking member **78**, and the pivot member **80** is also pivotable with respect to the button **82**.
[0040] In the illustrated embodiment, the locking member **78** extends through aligned openings formed in the knob body **74**, the pivot member bracket **106**, the mounting bracket **104** and the housing **108**. Furthermore, the housing **108** is configured to slidably receive the locking portion **88** of the locking member **78** as the locking member **78** moves between the extended and retracted positions. The knob mechanism **20** also includes a biasing member disposed in an interior of the housing **108** and that is configured to urge or bias the locking member **78** toward the extended position. For example, the knob mechanism **20** may include a compression spring **118**, such as a helical or coil spring, positioned around the locking pin **84** of the locking member **78** so that the spring **118** extends between the locking portion **88** of the locking member **78** and an end of the housing **108** attached to the mounting bracket **104**.
[0041] The knob mechanism **20** may be assembled in any suitable manner. For example, referring to FIGS. **7**, **8** and **10**, the locking pin **84** may be inserted into the locking portion **88** and attached to the locking portion **88** with the fastener **90**. Next, the compression spring **118** may be positioned over the locking pin **84** and the locking pin **84** may be inserted through the end of the housing **108** so that the locking portion **88** is received in the housing **108**. That subassembly may then be mounted to one side of the mounting bracket **104** using the screws **114**. The pivot member **80** may be attached to the pivot member bracket **106** with the pivot pin **116**, and the pivot member bracket **106** may then be positioned within the knob body **74**. Next, the knob body **74** and pivot member bracket **106** may be attached to the other side of the mounting bracket **104** using the screws **110** and nuts **112**. As shown in FIG. **10**, the knob body **74** may have an open outer end that provides access to the nuts **112** or other fasteners. The knob mechanism **20** may also include a cap **120**, which may be considered part of the knob body **74**, for covering the open outer end after assembly of the rest of the knob mechanism **20**. The knob mechanism **20** may then be attached in any suitable manner to the lever arm **46** of the actuator **30**, or other component if the knob mechanism **20** is used in a different application. Referring to FIG. **9**, for example, the housing **108** of the knob mechanism **20** may be inserted into a receptacle **122** on the lever arm **46**, and the housing **108** may be attached to the receptacle **122** using one or more fasteners **124**, such as a set screw.
[0042] Referring to FIGS. **1-9**, and as mentioned above, the knob mechanism **20** may be operable to lock the lever arm **46** in one or more desired positions during operation of the lift **10**. For example, an operator of the lift **10** may press the button **82** of the knob mechanism **20** inwardly in order to pivot the pivot member **80** and move the locking member **78** from the extended position shown in FIG. **7** to the retracted position shown in FIG. **8** in order to unlock the lever arm **46** from

the locking plate **70**. The lever arm **46** may then be rotated in a first direction (e.g., clockwise in the embodiment shown in FIGS. **1** and **5**) to thereby raise the support member arrangement **14** from the lowered position to the raised position, or a position between the lowered position and the raised position. For example, referring to FIGS. **1-5**, when the operator rotates the lever arm **46** in the first direction, such as clockwise in the embodiment shown in FIGS. **1** and **5**, the transmission **47** will cause the second drive member **28***b* to move, e.g., rotate or circulate, in a second direction opposite the first direction, such as counterclockwise in the embodiment shown in FIGS. **3**A and **4**. As a result, the third upright support member **26***c* will move upwardly because of the fixed interconnection between the second drive member **28***b* and the second upright support member **26***b*. Likewise, the upward movement of the third upright support member **26***c* will also cause the second upright support member **26***b* to move upwardly because of the fixed interconnection between the third upright support member **26***c* and the first drive member **28***a* (which will move, e.g., rotate or circulate, in the first direction, e.g., clockwise in the embodiment shown in FIGS. **3**A and **4**) and the fixed interconnection between the first drive member **28***a* and the first upright support member **26***a*. Specifically, because of the fixed interconnection between the third upright support member **26***c* and the first drive member **28***a*, upward movement of the third upright support member **26***c* will cause the first drive member **28***a* to move, e.g., rotate or circulate, in the first direction, e.g., clockwise in the embodiment shown in FIGS. **3**A and **4**, and that movement will cause the second upright support member **26***b* to move upwardly because of the fixed interconnection between the first drive member **28***a* and the first upright support member **26***a*. The second actuator **45** may also assist in upward movement of the second and third upright support members **26***b* and **26***c*, respectively, by urging the second support member **26***b* upwardly due to the arrangement of the second actuator **45** between the base **12** and the second upright support member **26***b*.

[0043] Referring to FIGS. **7-9**, when a desired elevation of the support member arrangement **14** and/or the work platform assembly **18** has been achieved, the operator may stop rotating the lever arm **46** so that movement of the second drive member **28***b* will stop, and the operator may release the button **82** of the knob mechanism **20** so that the locking member **78** may move toward the extended position and the button **82** may move toward the first position until the locking portion **88** of the knob mechanism **20** is able to be received in one of the locking apertures **72** of the locking plate **70** to thereby lock the lever arm **46** with respect to the locking plate **70**. When it is desired to lower the support member arrangement **14** and the work platform assembly **18**, the operator may again grasp the knob mechanism **20** and press the button **82** inwardly in order to unlock the lever arm **46** from the locking plate **70**. The lever arm **46** may then be rotated in the second direction (e.g., counterclockwise in the embodiment shown in FIGS. **2** and **5**) opposite the first direction to thereby lower the support member arrangement **14** to or toward the lowered position. Such movement of the lever arm **46** will cause the second drive member **28***b* to move, e.g. rotate or circulate, in the first direction, e.g., clockwise in the embodiment shown in FIGS. **3**A and **4**. As a result, the third upright support member **26***c* will move downwardly because of the fixed interconnection between the second drive member **28***b* and the second upright support member **26***b*. Likewise, the downward movement of the third upright support member **26***c* will also cause the second upright support member **26***b* to move downwardly because of the fixed interconnection between the third upright support member **26***c* and the first drive member **28***a* (which will move, e.g., rotate or circulate, in the second direction, e.g., counter-clockwise in the embodiment shown in FIGS. **3**A and **4**) and the fixed interconnection between the first drive member **28***a* and the first upright support member **26***a*. The button **82** of the knob mechanism **20** may then be released in order to lock the support member arrangement **14** in the lowered position, or any position between the raised position and the lowered position.

[0044] The above configuration of the knob mechanism **20** may provide numerous advantages or benefits. For example, the knob mechanism **20** may allow efficient actuation of the locking member **78** when the operator presses the button **82**. Furthermore, when the operator releases the button **82**, the spring **118** or other biasing member may urge the locking member **78** toward the extended position to thereby lock the lever arm **46** with respect to the locking plate **70** in order to inhibit or prevent further movement of the support member arrangement **14** and the work platform assembly **18**. In addition, the configuration of the knob mechanism **20** may enable easy assembly and disassembly.

[0045] FIGS. **11-14** show another embodiment **20**′ of a knob mechanism (e.g., handle mechanism) according to the disclosure, which may be used in any of the applications described above with respect to the knob mechanism **20**. The knob mechanism **20**′ is similar to the knob mechanism **20**, and similar features are identified with the same or similar reference numerals, except the reference numerals for the knob mechanism **20**′ may each include a prime mark. Therefore, the below description will primarily focus on features of the knob mechanism **20**′ that are different than the knob mechanism **20**. For example, with the knob mechanism **20**′, knob body **74**′ is connected directly to housing **108**′ with one or more fasteners, such as screws, bolts, nuts, etc., so the mounting bracket **104** from the knob mechanism **20** is not needed. In the illustrated embodiment, the knob body **74**′ is connected to the housing **108**′ with two screws **114**′ and corresponding washers **125**, for example. Like the knob body **74** of the knob mechanism **20**, the knob body **74**′ may have an open outer end that provides access to the screws **114**′ or other fasteners. The knob mechanism **20**′ may also include a cap **120**′, which may be considered part of the knob body **74**′, for covering the open outer end after assembly of the rest of the knob mechanism **20**′. The cap **120**′ may be attached to a main portion of the knob body **74**′ with one or more fasteners, such as screws **126**.

[0046] As another example, the knob mechanism **20**′ includes a locking member **78**′ having a locking pin **84**′ formed as two pieces that are joined together. Specifically, the locking pin **84**′ includes a first or main portion, such as locking pin portion **84***a*′, and a second portion, such as head portion **84***b*′, that is fixedly connected to the locking pin portion **84***a*′ in any suitable manner, such as by a threaded connection. Alternatively, the locking pin **84**′ may be formed as one piece.

[0047] As yet another example, the knob mechanism **20**′ includes a button **82**′ that is pivotally connected to the knob body **74**′ in any suitable manner, and a pivot member **80**′ that is fixedly connected to the button **82**′ in any suitable manner. In the illustrated embodiment, for example, the button **82**′ is pivotally connected to the knob body **74**′ with one or more fasteners, such as a pivot pin **127**, and the pivot member **80**′ is fixedly connected to the button **82**′ with one or more fasteners, such as a screw **128**. In another embodiment, the pivot member **80**′ may be integrally formed with the button **82**′ (e.g., the pivot member **80**′and the button **82**′ may be formed as one piece). While the pivot member **80**′ may have any suitable configuration, in the illustrated embodiment the pivot member **80**′ has a generally Z-shaped configuration with a first flat portion **130** attached to the button **82**′ so that an edge of the first flat portion **130** abuts a lip or projection **132** formed on a button body of the button **82**′. The pivot member **80**′ further has a second flat portion **134** that extends transversely to the first flat portion **130**, and a third flat portion **136** that extends transversely to the second flat portion **134** and generally parallel to the first flat portion **130**. The second and third flat portions **134** and **136**, respectively, also define an opening for receiving the head portion **84***b*′ of the locking member **78**′.

[0048] As with the knob mechanism **20**, the button **82**′ of the knob mechanism **20**′ is operable to pivot the pivot member **80**′ to move the locking member **78**′ from a first or extended position (e.g., locked position or non-actuated position) shown in FIG. **13** toward a second or retracted position (e.g., released position or actuated position) shown in FIG. **14**. Specifically, the pivot member **80**′ is pivotable with the button **82**′ as the button **82**′ pivots between a first position shown in FIG. **13** and a second position shown in FIG. **14**. The pivot member **80**′ is also engageable with the head portion **84***b*′ of the locking member **78**′ in order to move the locking member **78**′ from the extended position to the retracted position. For example, the third flat portion **136** of the pivot member **80**′

may be directly engageable with an enlarged head **94**′ of the head portion **84***b*′.
[0049] FIGS. **15-17** show yet another embodiment **20**″ of a knob mechanism (e.g., handle mechanism) according to the disclosure, which may be used in any of the applications described above with respect to the knob mechanism **20**. The knob mechanism **20**″ is generally similar to the knob mechanism **20**′, and similar features are identified with the same or similar reference numerals, except the reference numerals for the knob mechanism **20**″ may each include a double prime mark. Therefore, the below description will primarily focus on features of the knob mechanism **20**″ that are different than the knob mechanism **20**′. For example, the knob mechanism **20**″ may include a knob body **74**″ having a different configuration than the knob body **74**′ of the knob mechanism **20**′. In that regard, the knob body **74**″ may include a main portion having a smaller diameter than a corresponding main portion of the knob body **74**′ so that the knob body **74**″ may be joined to housing **108**″ with one or more fasteners, such as screws **114**″, that are accessible from an exterior of the knob body **74**″. The knob body **74**″ may also have a larger length than the corresponding main portion of the knob body **74**′. Likewise, locking member **78**″ of the knob mechanism **20**″ may include first portion, such as locking pin portion **84***a*″, and/or a second portion, such as head portion **84***b*″, that are longer than the respective locking pin portion **84***a*′ and head portion **84***b*′ of the locking member **78**′.
[0050] The knob mechanism **20**′ and the knob mechanism **20**″ may provide the same or similar advantages or benefits as described above with respect to the knob mechanism **20**.
[0051] The following clauses describe aspects of embodiments according to the disclosure.
[0052] Clause 1. A knob mechanism according to the disclosure may include a knob body and a lock mechanism associated with the knob body. The lock mechanism may include a locking member that is movable between an extended position and a retracted position, a pivot member that is pivotable with respect to the knob body to move the locking member, and a button that is operable to pivot the pivot member to move the locking member from the extended position toward the retracted position.
[0053] Clause 2. The knob mechanism of the preceding clause wherein the pivot member is engageable with the locking member to move the locking member from the extended position toward the retracted position.
[0054] Clause 3. The knob mechanism of any of the preceding clauses wherein the pivot member has a curved end that is engageable with the locking member when the locking member is in the retracted position.
[0055] Clause 4. The knob mechanism of any of the preceding clauses wherein the button is engageable with the pivot member to pivot the pivot member.
[0056] Clause 5. The knob mechanism of any of the preceding clauses wherein the pivot member has a curved end that is engageable with the button when the locking member is in the extended position.
[0057] Clause 6. The knob mechanism of any of the preceding clauses wherein the locking member is movable in a first direction, and the button is movable in a second direction transverse to the first direction.
[0058] Clause 7. The knob mechanism of clause 6 wherein the second direction is perpendicular to the first direction.
[0059] Clause 8. The knob mechanism of clause 6 wherein the pivot member has a longitudinal axis that extends at an angle greater than 0° with respect to the second direction when the locking member is in the extended position.
[0060] Clause 9. The knob mechanism of any of clauses 6-9 wherein the pivot member has a longitudinal axis that extends at an angle greater than 0° with respect to the first direction when the locking member is in the retracted position.
[0061] Clause 10. The knob mechanism of any of clauses 1-4 wherein the button is pivotally attached to the knob body.

[0062] Clause 11. The knob mechanism of clause 10 wherein the pivot member is fixedly attached to the button.
[0063] Clause 12. The knob mechanism of any preceding clause wherein the pivot member has an opening through which the locking member extends.
[0064] Clause 13. The knob mechanism of any preceding clause wherein the locking member comprises an elongated locking pin and an enlarged locking portion attached to the locking pin.
[0065] Clause 14. The knob mechanism of clause 13 wherein the locking portion is attached to the locking pin with a fastener that extends in a direction transverse to a longitudinal axis of the locking pin.
[0066] Clause 15. The knob mechanism of any of clauses 1-10 and 12-14 wherein the pivot member is pivotable about a pivot axis that extends between the button and the locking member.
[0067] Clause 16. The knob mechanism of any of clauses 1-10 and 12-15 further comprising a bracket disposed in an interior of the knob body, and the pivot member is pivotally attached to the bracket.
[0068] Clause 17. The knob mechanism of any preceding clause further comprising a housing attached to the knob body, and a biasing member disposed in an interior of the housing and configured to bias the locking member toward the extended position.
[0069] Clause 18. The knob mechanism of clause 17 wherein the biasing member comprises a compression spring.
[0070] Clause 19. The knob mechanism of any of clauses 17-18 further comprising a mounting bracket disposed between the housing and the knob body, wherein the locking member extends through the mounting bracket.
[0071] Clause 20. The knob mechanism of any of clauses 17-18 wherein the housing is attached to the knob body with one or more fasteners that are accessible from an interior of the knob body.
[0072] Clause 21. A lift device comprising a work platform; a movable support member arrangement that supports the work platform; and a drive system configured to move the support member arrangement in order to move the work platform, wherein the drive system includes an actuator that comprises the knob mechanism of any preceding clause.
[0073] Clause 22. The lift device of clause 21 wherein the actuator comprises a rotatable member, and the knob mechanism is mounted on the rotatable member.
[0074] Clause 23. The lift device of clause 22 further comprising a locking member mounted on the work platform or the support member arrangement, wherein the locking member of the knob mechanism is engageable with the locking member on the work platform or the support member arrangement to lock the rotatable member with respect to the work platform or the support member arrangement.
[0075] Clause 24. Any of the preceding clauses 1-23 in any combination.
[0076] While exemplary embodiments are described above, it is not intended that these embodiments describe all possible forms according to the disclosure. In that regard, the words used in the specification are words of description rather than limitation, and it is understood that various changes may be made without departing from the spirit and scope of the disclosure. Additionally, the features of various implementing embodiments may be combined to form further embodiments according to the disclosure. For example, the features of any one of the knob mechanism embodiments described above may be incorporated into any of the other above described embodiments.

Claims

1. A knob mechanism comprising: a knob body; and a lock mechanism associated with the knob body, the lock mechanism comprising a locking member that is movable between an extended position and a retracted position, a pivot member that is pivotable with respect to the knob body to move the locking member, and a button that is operable to pivot the pivot member to move the locking member from the extended position toward the retracted position.

2. The knob mechanism of claim 1 wherein the pivot member is engageable with the locking member to move the locking member from the extended position toward the retracted position.

3. The knob mechanism of claim 2 wherein the pivot member has a curved end that is engageable with the locking member when the locking member is in the retracted position.

4. The knob mechanism of claim 2 wherein the button is engageable with the pivot member to pivot the pivot member.

5. The knob mechanism of claim 4 wherein the pivot member has a curved end that is engageable with the button when the locking member is in the extended position.

6. The knob mechanism of claim 1 wherein the locking member is movable in a first direction, and the button is movable in a second direction transverse to the first direction.

7. The knob mechanism of claim 6 wherein the pivot member has a longitudinal axis that extends at an angle greater than 0° with respect to the second direction when the locking member is in the extended position.

8. The knob mechanism of claim 1 wherein the button is pivotally attached to the knob body.

9. The knob mechanism of claim 1 wherein the pivot member is fixedly attached to the button.

10. The knob mechanism of claim 9 wherein the pivot member is integrally formed with the button.

11. The knob mechanism of claim 1 wherein the pivot member has an opening through which the locking member extends.

12. The knob mechanism of claim 1 wherein the locking member comprises an elongated locking pin and an enlarged locking portion attached to the locking pin.

13. The knob mechanism of claim 1 wherein the pivot member is pivotable about a pivot axis that extends between the button and the locking member.

14. The knob mechanism of claim 1 further comprising a housing attached to the knob body, and a biasing member disposed in an interior of the housing and configured to bias the locking member toward the extended position.

15. The knob mechanism of claim 14 wherein the biasing member comprises a compression spring.

16. The knob mechanism of claim 14 wherein the housing is attached to the knob body with one or more fasteners that are accessible from an interior of the knob body.

17. The knob mechanism of claim 14 wherein the housing is attached to the knob body with one or more fasteners that are accessible from an exterior of the knob body.

18. A lift device comprising: a work platform; a movable support member arrangement that supports the work platform; and a drive system configured to move the support member arrangement in order to move the work platform, the drive system including an actuator that comprises the knob mechanism of claim 1.

19. The lift device of claim 18 wherein the actuator comprises a rotatable member, and the knob mechanism is mounted on the rotatable member.

20. The lift device of claim 19 further comprising a locking member mounted on the work platform or the support member arrangement, wherein the locking member of the knob mechanism is engageable with the locking member on the work platform or the support member arrangement to lock the rotatable member with respect to the work platform or the support member arrangement.